

Trainings ▾

General Information

⚠ CAUTION ⚠

The plumbing and make-up tank included in the Centinel™ kits may not be adequate for use in all applications. All applicable local, state and federal regulations and any institutional codes must be consulted to ensure that all components satisfy these regulations and codes. Any component that does not satisfy any applicable regulation or code must be replaced with a component that does.

Note : Refer to Procedure 208-002 (</qs3/pubsys2/xml/en/procedures/96/96-208-002.html>) for appropriate tank mounting location.

The small and large make-up oil tanks **must** be supported by at least four bolts. If **only** four bolts are used, they **must** be located on all four corners of the make-up oil tank.

To help prevent leaks, apply pipe sealing compound to any pipe fittings.

There is no make-up oil tank to compensate for oil burned via the Centinel™ burn-only system. Oil **must** be added to the engine on a daily basis to compensate for oil burned via the Centinel™ system and to compensate for normal engine oil consumption.

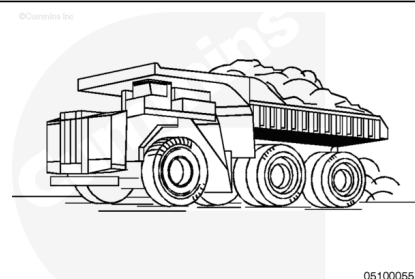
Vehicles domiciled in California can **not** use any system (like Centinel™) that blends used engine oil with diesel fuel. (Check CARB regulations.)

Note : Marine applications require that all hoses and fittings comply to SAE standards J1942 and J1527. **Not** all hoses and fittings supplied in the Centinel™ kits comply with these SAE standards. A metal make-up tank can also be required for Marine applications.

Install

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components satisfy these regulations and codes. Any component that does not satisfy any applicable regulation or code must be replaced with a component that does.

Note : The area selected must be convenient for the operator to fill the make-up oil tank and see the fault code warning lamps.

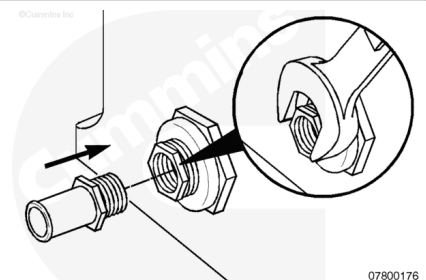
Mount the make-up oil tank assembly.

Note : The following torque requirements are for plastic tank fasteners.

- Maximum torque when using self-locking nuts is 34 N•m [25 ft-lb].
- Maximum torque when **not** using self-locking nuts is 20 N•m [177 in-lb].
- **Always** use a 3/4-inch outer diameter round washer (maximum) with the old style tank design. Currently, this **only** applies to the 24.6 liters [6.5-gallon] tank.
- **Always** use a 7/8-inch square washer with the new style tank design. Currently, this **only** applies to the 15-liter [4-gallon] tanks.

⚠ CAUTION ⚠

When installing the fitting or elbow into the make-up tank, hold the make-up tank fitting with a wrench while tightening the fitting or elbow. Failure to hold the make-up tank fitting during installation of the fitting or elbow can cause damage to the bottom of the tank.

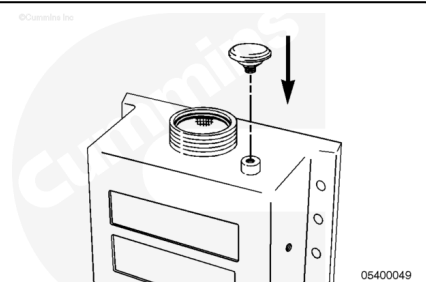


Install the 3/4-inch pipe thread fitting to the bottom of the make-up tank.

Torque Value: 7 n•m [62 in-lb]

Install the vent to the top of the make-up tank.

Torque Value: 3 n•m [27 in-lb]

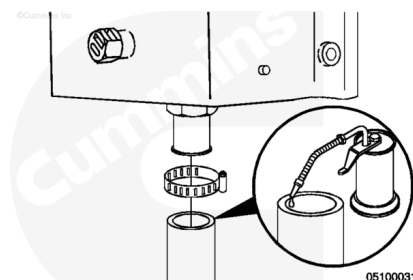


Note : Use engine lubricating oil to lubricate each end of the hoses for easier installation.

Locate the 3/4-inch hose in the kit.

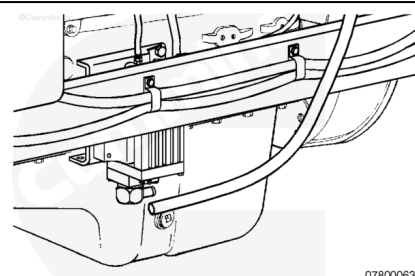
Install a hose clamp onto the hose and put the hose on the drain of the oil make-up tank.

Slide the hose clamp into position and tighten. Refer to the manufacturer's torque value.



⚠ WARNING ⚠

Hoses must not be routed near excessively hot components and must be securely fastened to be sure they do not contact hot or moving components during any operating conditions. If routing near hot components can not be avoided, then a fire sleeve around the hose must be used. All hoses must be adequately protected from chafing and fastened to appropriate structure, such that there is no overhung load on connections.



Route the 3/4-inch clean oil supply hose from the make-up tank to the Centinel valve. The hose **must not** be routed near any moving parts or hot surfaces.

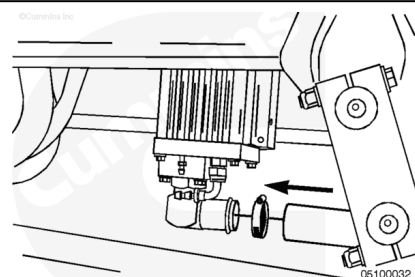
Cut the hose to fit.

For easier installation, use engine oil to lubricate the end of the hose before installing the hose onto the drain fitting.

Slide the appropriate hose clamp onto the hose and install the hose onto the oil control valve make-up oil supply port.

Slide the hose clamp into position and tighten. Refer to the manufacturer's torque value.

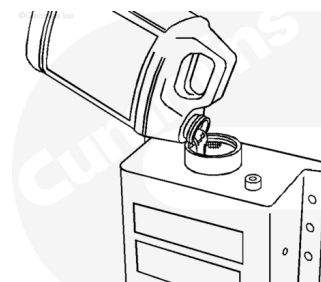
Using the provided p-clamps and tie wraps, secure the hose to the frame rail.



Prime

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



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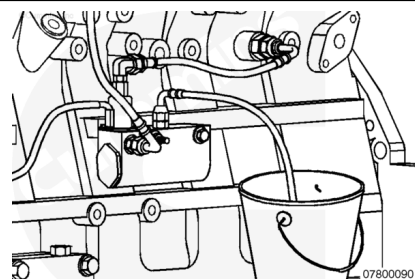
⚠ CAUTION ⚠

If the system is not properly primed, improper system operation can occur, resulting in make-up oil not being added to the engine.

Fill the make-up tank with clean engine oil.

Check for leaks.

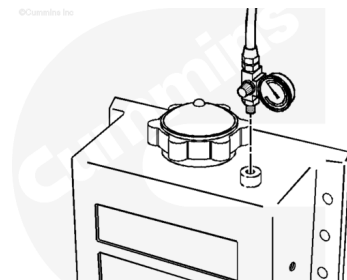
Disconnect the oil replenishment hose from the dipstick port hole and route it into a bucket or other oil containment item.



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⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



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⚠ CAUTION ⚠

When applying air pressure to the make-up tank, do not exceed 124 kPa [18 psi] air pressure. Exceeding 124 kPa [18 psi] air pressure will cause damage to the make-up tank.

Temporarily plug any tank vents as necessary and apply 69 to 103 kPa [10 to 15 psi] air pressure to the oil make-up tank to force the oil through the hose and valve. Due to installation variances, this pressure will probably need to be increased in 14 to 21 kPa [2 to 3 psi] increments.

Note : This purging pressure requirement must be accounted for in any non-Cummins-supplied oil make-up tank design.

Monitor the disconnected fresh oil hose for oil flow.

Once oil flows out the hose, the system has been primed.

Disconnect the shop air supply from the make-up tank and connect the hose to the dipstick port.

Unplug the vents, and replace the vent and fill cap.

Refill the oil make-up tank with clean engine oil to the FULL mark on the tank.

